WORKSHEET

Chapter 10: Sample sizes

1 If the sample proportion (p) is approximately 0.7, what sample size should be used to estimate p with an error margin of 5% with 98% confidence?

2 The Department of Justice wants to estimate the proportion of serious crimes that involve violence with an error margin of 2% with 90% confidence. Data from previous years shows that the proportion is approximately 0.6. What sample size should be taken?

3 An ice cream manufacturer wants to estimate the proportion of people who would eat sarsaparilla ice cream with an error margin of 3% with 90% confidence. It is estimated that that p is about 0.2. How many people should be surveyed?

4 The probability of a particular property in a population is unknown. What sample size should be used to establish the probability with an error margin of 5% at a confidence level of 95%?

5 If we require a 90% confidence interval for p with an error margin of 3%, what sample size should be used?

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6 How many births would need to be checked to establish the true proportion of a newborn baby being a girl in Australia to within 5% at a confidence level of 98%?

7 The World Health Organization determined hepatitis B to be the fourth biggest killer among the world's infectious diseases. We want to take a sample of the citizens of a particular city to determine the percentage of people who have, or are carrying, hepatitis B. How many people should be in the sample for a 95% confidence interval to within 5%?

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Answers

1 About 500 (455)

2 About 1600 (1624)

3 About 500 (482)

4 About 400 (385)

5 About 800 (752)

6 About 500 (543)

7 About 1500 (1537)

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